Using of J. Young type solvent flask

Date: 2024-09-09
Category: Protocol
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Goal

This protocol should highlight the important tricks, when using J. Young solvent flask.

There are two different sizes: 500 ml (big ones) are used for dry solvents and the 25 ml (small ones) are used for dry NMR solvents.

Filling the J. Young flask

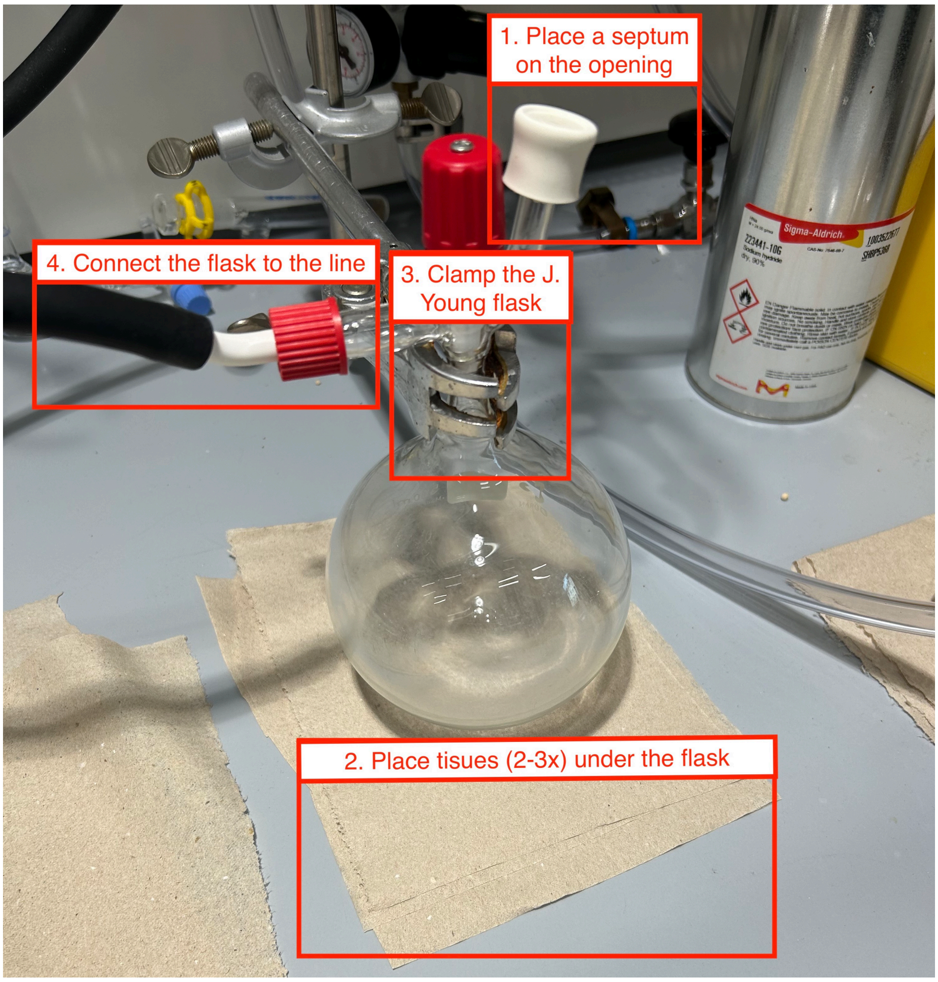
If the solvent is taken out of the SPS: Protocol - Collecting of SPS solvents

Degassing, when using this type of flasks: Protocol - Degassing of organic solvents in Young-type flask

Setup

First steps to take, when using J. Young type flasks.

Step number	Step description
1	Place a septum on the opening
2	Place tissues under the flask
3	Clamp the flask under the "Y-thing"
4	Connect the flask to the line using GL14



Taking solvent out of the J. Young flask using a syringe

Here, the way without touching/damaging the PTFE valve is described. Important: the tip needle should not touch the PTFE valve!

There is a video of the whole process: whole_process_01-05.mov

And two videos of the steps 06-07, where the movement of how NOT to touch the PFTE valve is shown precisely: steps_04_05.mov, steps_04_05_other_angle.mov

Step number	Step description
Preparation	See "set up" above setup.jpeg and picture preparation.jpeg
01	Preparation of the needle, the needle is bend at the top part (ca. 1 cm from the end) in ca. 35 °. For this always use a tissues, since you don't know what is on your gloves. 01_needle.jpeg, 01_needle_closer.jpeg

02	After three cycles of changing the atmosphere in the needle, place the needle through the septum. 02_going_in.jpeg Go with the needle up to the PTFE valve, but still keep a distance. *Important: the tip needle should not touch the PTFE valve!* 02_1_first_stop.jpeg
03	Open the PTFE valve up to the point where it is fully open, but the ring (bottom part) stays on. 03_opening.jpeg
04	Go through the opening next the the PTFE valve in a way that the tip of the needle is sliding above the glas and pointed downwards. *(Better seen in the video)* 04_going_through.jpeg Video: steps_04_05.mov, steps_04_05_other_angle.mov
05	Now just push the needle as much as you can. 05_you_are_in.jpeg Video: steps_04_05.mov, steps_04_05_other_angle.mov
06	Turn the needle by 180 ° and you should be able to take of the solvent. For this the flask needs to be clamped out and tilted as much as you need to. 06_taking_the_solvent.jpeg *Since these are solvents, you should omit putting the solution back to the solvent flask and just take everything out what was once in your needle.*
07	Take the needle above the "Y"-part and bend the needle using your fingers. *Never bend needles using the septum.*
08	Close the PTFE valve.
09	Take an argon puffer on top of your solution in the needle and take it out.
10	Congratulation, you are done :)

Linked resources

Protocol - Degassing of organic solvents in Young-type flask

Protocol - Collecting of SPS solvents

Attached files

setup.jpeg
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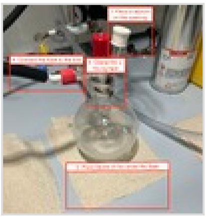

whole_process_01-05.mov

steps_04_05.mov
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steps_04_05_other_angle.mov
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04_going_through.jpeg
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05_you_are_in.jpeg
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06_taking_the_solvent.jpeg
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01_needle_closer.jpeg
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02_going_in.jpeg
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02_1_first_stop.jpeg
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03_opening.jpeg
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preparation.jpeg
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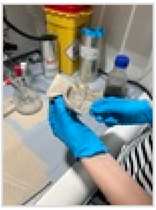

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